The Omega Automath: Finite Holographic, Inductive-Colimit, and Mirror-Phase Corridors in Lean 4

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Abstract

This manuscript records a theorem-safe synthesis of the finite Lean 4 and SymPy artifacts currently present in the `info-geometry-lean` repository. The verified core is a family of finite algebraic corridors: Cuntz/UHF horizon nilpotence and reconstruction identities, scalar Bekenstein–Hawking/dyadic entropy calibrations under explicit nonzero-denominator hypotheses, common-phase modular-flow crossing invariance, an inductive-colimit crystal readout for the $Cl(1,1)$ tensor tower, and a finite two-branch attention projection.

The document intentionally does not claim a proof of full AdS/CFT, the Black Hole Information Paradox, HKLL in its analytic continuum form, the Standard Model Higgs mechanism, the Riemann Hypothesis, the Mass Gap theorem, or hallucination-freedom of arbitrary large language models. Instead, it records what is kernel-checked: explicit finite identities and conditional bridges that can serve as stable atoms for future continuum or physical hypotheses.

1 Lean and SymPy Authority

All mathematical claims in this manuscript refer to compiled Lean owner modules and matching SymPy witnesses. The relevant Lean modules include:

- `InfoGeometry.Holography.AdSCFTCuntzBridge`;
- `InfoGeometry.Holography.BekensteinHawkingDyadicEntropy`;
- `InfoGeometry.Holography.WittenMobiusBekensteinComplement`;
- `InfoGeometry.Holography.ModularFlowKMS`;
- `InfoGeometry.Clifford.InductiveColimitCrystal`;
- `InfoGeometry.LLM.MirrorPhaseCuntzAttention`;
- `InfoGeometry.Canonical.DrazinModularPersistence`.

SymPy scripts mirror these finite reductions symbolically; they are evidence and regression witnesses, not substitutes for Lean proof.

2 Epoch 3: Finite Holographic Algebra

2.1 Nilpotent event-horizon readout

In `AdSCFTCuntzBridge`, the event-horizon toy operator is a finite Cuntz/UHF expression. Lean proves the nilpotence identity

$$N^2 = 0, \tag{1}$$

represented by the declaration `horizon_nilpotence`. It also proves the finite reconstruction identity

$$(NN^* + N^*N)X = X, \tag{2}$$

represented by `information_preservation`. This is a finite algebraic identity, not a continuum black-hole information theorem.

2.2 Scalar Bekenstein–Hawking/dyadic calibration

The scalar readout

$$S_{\mathrm{BH}}(A, G) = \frac{A}{4G} \tag{3}$$

is formalized in `BekensteinHawkingDyadicEntropy`. Under explicit calibration assumptions such as

$$A = 4G \log 2 \tag{4}$$

and the guardrail

$$G \neq 0, \tag{5}$$

Lean proves equality with the dyadic entropy quantum and with the two-branch Massieu readout. The nonzero hypothesis is a denominator condition required for field simplification and cancellation; it is physically suggestive but not by itself a theorem of emergent gravity.

2.3 Finite Witten–Möbius/Cuntz/Bekenstein complement

The module `WittenMobiusBekensteinComplement` bundles a finite certificate:

1. finite Möbius/parity agreement for a prime-bit state;
2. finite Witten cancellation over a finite powerset;
3. Cuntz horizon nilpotence;
4. Cuntz reconstruction on an arbitrary readout X;
5. scalar Bekenstein–Hawking calibration under the same explicit area and nonzero-G hypotheses.

This is a finite complement theorem, not an infinite arithmetic or physical closure theorem.

3 Common-Phase Modular Crossing

The module `ModularFlowKMS` isolates a purely algebraic KMS-shaped calculation. Suppose two elements transform by the same central unitary phase under a flow σ_t. Then Lean proves

$$\sigma_t(XY^*) = XY^*. \tag{6}$$

The symmetric crossing

$$S_L S_R^* + S_R S_L^* \tag{7}$$

is fixed because both summands are fixed. The theorem surface is finite and conditional. It does not construct an unbounded Tomita operator, prove analytic KMS boundary conditions, or identify the Standard Model Higgs field.

4 The Inductive-Colimit Crystal

The finite-to-infinite substrate is formalized in the $Cl(1,1)$ tensor-tower modules and summarized in `InductiveColimitCrystal`. The finite atom at stage n is denoted `AtomStage n`. The algebraic crystal carrier is the repository-owned direct limit `CrystalCarrier`. Finite binding is the stage advance

$$A \longmapsto A \otimes I_2, \tag{8}$$

abstracted by `bindAtoms`.

Lean proves that finite binding does not change the represented crystal element:

$$\iota_{m+k}(\mathrm{bind}_{m,k}(A)) = \iota_m(A). \tag{9}$$

It also proves transport of square-zero atoms, idempotent atoms, and inner commutator readouts, together with stability of normalized Markov trace and normalized log-determinant readouts.

The SymPy witness `inductive_colimit_crystal.py` checks the concrete Kronecker model:

$$\mathrm{tr}_{\mathrm{norm}}(A \otimes I_2) = \mathrm{tr}_{\mathrm{norm}}(A), \tag{10}$$

$$\det(A \otimes I_2) = \det(A)^2, \tag{11}$$

and verifies commutator, idempotent, and square-zero transport.

Remark 4.1. *The language of atoms binding into a crystal is mathematically apt for this algebraic direct-limit construction. Continuum Bloch waves, Klein-bottle Brillouin zones, phonons, and Virasoro continuum excitations remain additional structures requiring explicit hypotheses and owner theorems.*

5 Epoch 4: Mirror-Phase Finite Attention

The finite attention corridor lives in `MirrorPhaseCuntzAttention`. The exact two-branch attention matrix is

$$A = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \tag{12}$$

Lean proves:

$$A^2 = A, \tag{13}$$

$$\sum_j A_{ij} = 1, \tag{14}$$

$$A_{ij} \geq 0. \tag{15}$$

For a two-branch vector $v = (v_0, v_1)$, applying attention gives

$$Av = \left(\frac{v_0 + v_1}{2}, \frac{v_0 + v_1}{2} \right). \tag{16}$$

Thus the branch anomaly $v_1 - v_0$ is killed, total branch mass is preserved, and fixed points are exactly branch-balanced vectors.

Remark 5.1. *This is a genuine finite projection theorem. It is not a theorem that arbitrary trained transformer attention is Cuntz-exact, that semantic hallucination is impossible, or that repeated LLM inference is thermodynamically lossless.*

6 Repaired Proof Debt

Two proof-debt repairs are part of the current theorem-safe surface:

- In `DrazinModularPersistence`, positivity of a real expectation state is now the non-vacuous predicate

$$\forall A, \quad 0 \leq \varphi(A^* A). \tag{17}$$

- In `OperatorAlgebra.SpectralTriple`, previously impossible renormalization socket claims were repaired by adding explicit finiteness, off-pole continuity, and cyclicity witnesses.

7 Validation Commands

Representative commands used to validate the surfaces include:

```
python3 tools/infra/run_locked_lake_build.py --wait-for-build-lock \
  InfoGeometry.Holography.ModularFlowKMS \
  InfoGeometry.Holography.BekensteinHawkingDyadicEntropy \
  InfoGeometry.Holography.WittenMobiusBekensteinComplement \
  InfoGeometry.Holography.All

python3 tools/infra/run_locked_lake_build.py --wait-for-build-lock \
  InfoGeometry.Clifford.InductiveColimitCrystal \
  InfoGeometry.Clifford.All

python3 tools/infra/run_locked_lake_build.py --wait-for-build-lock \
  InfoGeometry.LLM.MirrorPhaseCuntzAttention \
  InfoGeometry.LLM
```

Representative SymPy witnesses include:

```
python3 tools/sympy/holographic_kan_cuntz.py
python3 tools/sympy/bekenstein_hawking_dyadic_entropy.py
python3 tools/sympy/witten_mobius_bekenstein_complement.py
python3 tools/sympy/modular_flow_kms.py
python3 tools/sympy/inductive_colimit_crystal.py
python3 tools/sympy/mirror_phase_cuntz_attention.py
python3 tools/sympy/drazin_modular_persistence.py
```

8 Closure Debt

The following are research directions rather than current Lean theorems:

- continuum AdS/CFT or HKLL reconstruction;
- the physical Black Hole Information Paradox;
- full Bekenstein–Hawking derivation from a Cuntz boundary;
- unbounded Tomita–Takesaki theory and analytic KMS strips;
- Klein-bottle Brillouin-zone classification from the current colimit;
- Bloch-wave, phonon, or Virasoro continuum limits from the finite tower;

- Standard Model Higgs mass derivation;
- Riemann Hypothesis or Mass Gap;
- hallucination-freedom of trained large language models.

9 Conclusion

The current repository contains a robust finite theorem substrate: Cuntz/UHF nilpotence and reconstruction, dyadic entropy calibration, modular crossing invariance, algebraic direct-limit crystal transport, and finite attention projection. These results are small enough to be kernel-checked and concrete enough to serve as atoms for future continuum hypotheses. The next mathematically safe step is to add explicit Bloch/Klein/continuum hypotheses and prove only the properties that follow from them.